Web Dev Mastery

📘 CSS: Margin, Padding & Box Model

🔲 1. CSS Box Model Overview

The **CSS Box Model** describes how elements are displayed on a web page. Every HTML element is considered as a **box**, and it consists of:

```
+---------------------------+
|        Margin             |   ← Outside Space
|  +---------------------+  |
|  |     Border          |  |   ← Edge around Padding
|  |  +---------------+  |  |
|  |  |   Padding     |  |  |   ← Space inside Border
|  |  |  +----------+  |  |  |
|  |  |  |  Content  |  |  |  |   ← Actual text/image
|  |  |  +----------+  |  |  |
|  |  +---------------+  |  |
|  +---------------------+  |
+---------------------------+
```

🧱 2. Box Model Components

✅ a. Content

- The actual content like text, image, etc.

✅ b. Padding

- Space **inside** the element, between the content and border.

```
padding: 20px;  /* Adds space inside element */
```

✅ c. Border

- A border around the padding (optional).

```
border: 2px solid black;
```

✅ d. Margin

- Space **outside** the border. Used to separate elements.

```
margin: 20px;  /* Adds space outside element */
```

💻 3. Example: Box Model in Action

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .box {
      width: 200px;
      height: 100px;
      padding: 20px;
      border: 3px solid blue;
      margin: 30px;
      background-color: lightgray;
    }
  </style>
</head>
<body>

  <div class="box">
    This is a box.
  </div>

</body>
</html>
```

📏 4. Detailed Explanation of Margin and Padding

🔸 Padding Syntax

```
padding: 10px;              /* All sides */
padding: 10px 20px;         /* Top/Bottom = 10px, Left/Right = 20px */
padding: 10px 15px 20px;    /* Top = 10px, Left/Right = 15px, Bottom =
20px */
```

```
padding: 5px 10px 15px 20px;/* Top, Right, Bottom, Left */
```

🔸 **Margin Syntax**

```
margin: 10px;                /* All sides */
margin: 10px 20px;           /* Top/Bottom = 10px, Left/Right = 20px */
margin: 10px 15px 20px;      /* Top = 10px, Left/Right = 15px, Bottom =
20px */
margin: 5px 10px 15px 20px; /* Top, Right, Bottom, Left */
```

🧠 5. Important Notes

- `padding` increases the size **inside** the box.
- `margin` creates space **outside** the box.
- You can use `margin: auto;` to center elements horizontally.
- `box-sizing: border-box;` includes padding & border in width/height calculation.

🧪 6. Example with `box-sizing`

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .box {
      width: 200px;
      padding: 20px;
      border: 5px solid red;
      box-sizing: border-box;  /* Total width remains 200px */
      background-color: lightgreen;
    }
  </style>
</head>
<body>

  <div class="box">
    Box with box-sizing: border-box;
  </div>

</body>
</html>
```

✅ 7. Quick Summary Table

Property	Description	Affects Layout?
`margin`	Outside spacing	Yes
`padding`	Inside spacing	Yes
`border`	Outer edge around padding	Yes

`box-sizing`	Includes padding/border in width	Yes